

✓ Diffusion Policy for Robot Learning: A Pedagogical Guide

Welcome to this interactive demonstration of **Diffusion Policy**, a state-of-the-art approach for learning visuomotor policies from demonstrations.

Overview

This notebook will guide you through:

1. **Theoretical Foundations:** Understanding diffusion models and their application to imitation learning
2. **Environment Setup:** Installing dependencies and preparing the experimental environment
3. **Training:** Implementing and training a Diffusion Policy on the PushT manipulation task
4. **Evaluation:** Quantitative assessment of the learned policy
5. **Analysis:** Visualizing and interpreting the iterative denoising process

Introduction to Diffusion Policy

Diffusion Policy adapts the generative modeling framework of diffusion models (originally developed for image generation) to the domain of robotic action prediction:

Core Principle: Rather than directly regressing actions from observations, we model action generation as an iterative denoising process, starting from random Gaussian noise and gradually refining it into coherent action sequences.

Key Advantages:

- **Multimodality:** Naturally captures multiple valid solution strategies for a given task
- **Temporal Coherence:** Generates smooth, dynamically feasible action trajectories
- **Expressiveness:** Models complex, high-dimensional action distributions
- **Generalization:** Demonstrates improved robustness to distribution shift compared to standard behavioral cloning

The PushT Manipulation Task

We demonstrate the approach on **PushT**, a contact-rich manipulation benchmark:

- **Objective:** Manipulate a T-shaped rigid body to a target pose using a circular end-effector
- **State Space:** 2D planar environment with continuous state and action spaces
- **Challenges:** Non-prehensile manipulation, contact dynamics, precise pose control
- **Relevance:** Serves as a representative task for evaluating visuomotor policy learning algorithms

Part 1: Environment Setup

Install dependencies and configure the environment.

Theoretical Foundation: Diffusion Models for Action Generation

Forward Diffusion Process

The forward process gradually corrupts clean actions $\mathbf{a}_0 \in \mathbb{R}^{T_a \times d_a}$ (where T_a is the action horizon and d_a is action dimensionality) by adding Gaussian noise over T diffusion timesteps:

$$q(\mathbf{a}_t | \mathbf{a}_{t-1}) = \mathcal{N}(\mathbf{a}_t; \sqrt{1 - \beta_t} \mathbf{a}_{t-1}, \beta_t \mathbf{I})$$

where $\{\beta_t\}_{t=1}^T$ is a variance schedule. Using the reparameterization trick, we can directly sample at any timestep:

$$q(\mathbf{a}_t | \mathbf{a}_0) = \mathcal{N}(\mathbf{a}_t; \sqrt{\bar{\alpha}_t} \mathbf{a}_0, (1 - \bar{\alpha}_t) \mathbf{I})$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$.

Reverse Diffusion Process

The reverse process learns to denoise actions conditioned on observations $\mathbf{o}_t$:

$$p_\theta(\mathbf{a}_{t-1} | \mathbf{a}_t, \mathbf{o}) = \mathcal{N}(\mathbf{a}_{t-1}; \boldsymbol{\mu}_\theta(\mathbf{a}_t, \mathbf{o}, t), \boldsymbol{\Sigma}_\theta(\mathbf{a}_t, \mathbf{o}, t))$$

In practice, we predict the noise ϵ rather than the mean directly:

$$\mu_{\theta}(\mathbf{a}_t, \mathbf{o}, t) = \frac{1}{\sqrt{\alpha_t}} \left(\mathbf{a}_t - \frac{1 - \alpha_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_{\theta}(\mathbf{a}_t, \mathbf{o}, t) \right)$$

Training Objective

The model is trained to minimize the simplified denoising score matching objective:

$$\mathcal{L}(\theta) = \mathbb{E}_{t \sim \mathcal{U}(1, T), \mathbf{a}_0 \sim p_{\text{data}}, \epsilon \sim \mathcal{N}(0, \mathbf{I})} [\|\epsilon - \epsilon_{\theta}(\mathbf{a}_t, \mathbf{o}, t)\|^2]$$

where $\mathbf{a}_t = \sqrt{\bar{\alpha}_t} \mathbf{a}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$.

Why Diffusion for Robot Learning?

1. **Expressiveness:** The iterative refinement process can represent arbitrary continuous distributions, including multimodal action distributions arising from multiple valid strategies.
2. **Stability:** Unlike GANs, diffusion models have a stable training objective without adversarial dynamics. Compared to normalizing flows, they require simpler architectures.
3. **Temporal Consistency:** The denoising process naturally produces smooth trajectories in action space, critical for stable robot control.
4. **Sample Quality:** The iterative nature allows progressive refinement, producing high-fidelity action sequences that respect task constraints.

Network Architecture

System Overview

The complete pipeline consists of:

```

Observations: o_t = {I_t, s_t} (images + proprioceptive state)
    ↓
Vision Encoder: φ(I_t) → z_t ∈ R^d_z
    ↓
Observation Embedding: h_o = f_obs([z_t, s_t])
    ↓
Noisy Actions: a_t^(k) ~ N(θ, I) (initialization at k=T)
    ↓
    [ For k = T, T-1, ..., 1:
      | ε_θ(a_t^(k), h_o, k)
      | a_t^(k-1) = Denoise(a_t^(k), ε_θ)
    ]
    ↓
Clean Action Sequence: a_t^(0) ∈ R^(T_a × d_a)

```

Transformer-Based Denoiser

The noise prediction network ϵ_{θ} is implemented as a transformer:

$$\epsilon_{\theta}(\mathbf{a}_t, \mathbf{o}, t) = \text{Transformer}(\text{Embed}(\mathbf{a}_t) + \text{TimeEmbed}(t), \text{Embed}(\mathbf{o}))$$

Components:

- **Time Embedding:** Sinusoidal positional encoding of diffusion timestep t
- **Action Embedding:** Linear projection of noisy action sequence
- **Observation Encoding:** Vision features from CNN/ResNet backbone combined with proprioceptive state
- **Cross-Attention:** Attention mechanism between action queries and observation keys/values
- **Self-Attention:** Temporal modeling within the action sequence

Receding Horizon Action Execution

A critical design choice is the prediction-execution mismatch:

- **Prediction Horizon:** $T_p = 16$ actions
- **Execution Horizon:** $T_e = 8$ actions

At each control timestep:

1. Predict action chunk: $\hat{\mathbf{a}}_{t:t+T_p-1}$
2. Execute first T_e actions: $\mathbf{a}_{t:t+T_e-1}$
3. Advance by T_e steps and re-predict

Temporal Overlap:

```
Iteration 1: Predict [a1, a2, ..., a16] → Execute [a1, ..., a8]
Iteration 2:      Predict [a9, a10, ..., a24] → Execute [a9, ..., a16]
                ↑____overlap____↑
```

Rationale:

1. **Receding Horizon Control:** Similar to MPC, re-planning allows incorporation of new observations
2. **Temporal Smoothing:** Overlapping predictions create implicit consistency constraints
3. **Empirical Performance:** Chi et al. (2023) demonstrate significant improvement with $T_p > T_e$

Inference Algorithm

At test time, we sample actions using DDPM or DDIM sampling:

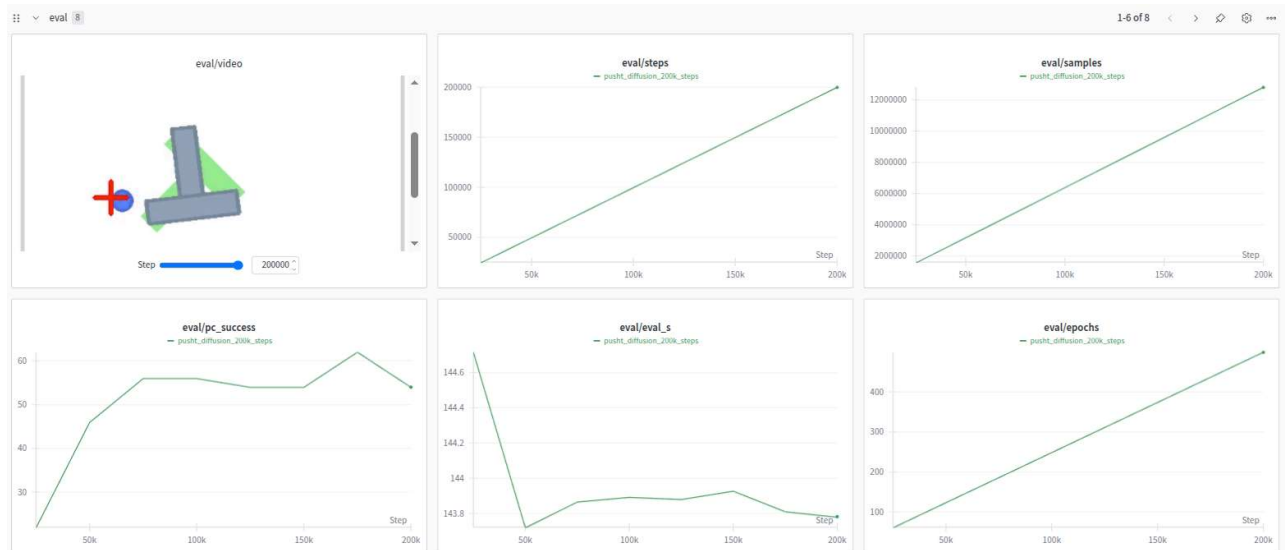
DDPM Sampling:

```
Initialize: aT^(T) ~ N(0, I)
For k = T down to 1:
    εpred = εθ(aT^(k), oT, k)
    aT^(k-1) = (1/√αk)[aT^(k) - (1-αk)/√(1-αk) · εpred] + σk · z
    where z ~ N(0, I)
Return: aT^(0)
```

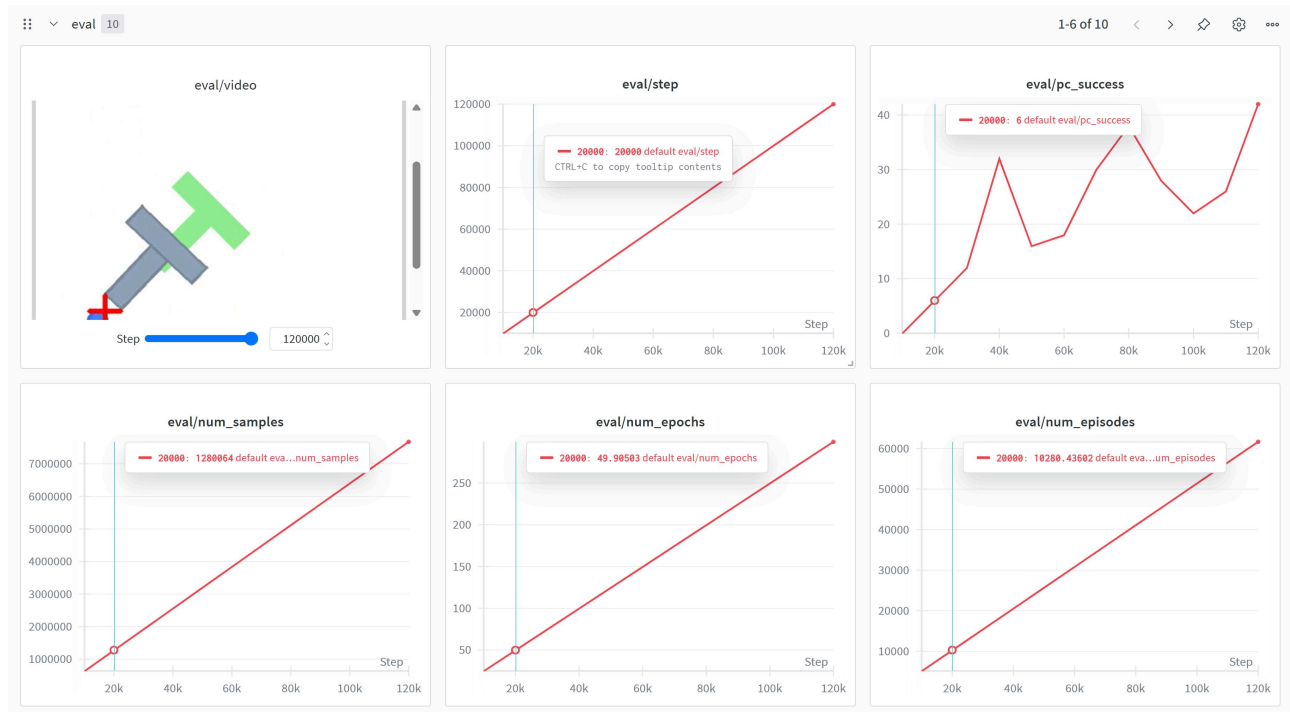
DDIM Sampling (deterministic, faster): Reduces sampling steps while maintaining quality through deterministic denoising trajectory.

Comparative Training runs of Transformer, Unet Diffusion Policy

1. Unet Based Diffusion Policy (~60% Success rate)



2. Transformer Based Diffusion Policy (~40% Success with 10% params)



*Note: Our two versions of original Diffusion Policy Successfully replicated the original results by showing a increase in success rate over training time.

```
# Demo of Transformer Based Denoiser
# Uncomment to train a transformer based Diffusion Policy

***Note**: The cell below trains a new model from scratch. This process is computationally intensive. For rapid experimentation, you

# %mkdir -p transformer_demo
# %cd transformer_demo
# !git clone -b add_transformer_diffusion_model https://github.com/amandip7/lerobot.git
# %cd lerobot
# !pip install .
# !pip install 'lerobot[pusht]' hydra-core
# %env WANDB_SERVICE=
# !PYTHONPATH=/content/lerobot python lerobot/scripts/train.py policy=diffusion_transformer env=pusht wandb.enable=false training.eval
# %cd ..
```

Continuing with standard U-net based Diffusion Policy - with Padagogical visualization

```
# Clone LeRobot repository with Unet-based diffusion model
!git clone -b gatech/reproduce_results https://github.com/kylekam/lerobot.git
```

```
Cloning into 'lerobot'...
remote: Enumerating objects: 13429, done.
remote: Counting objects: 100% (32/32), done.
remote: Compressing objects: 100% (14/14), done.
remote: Total 13429 (delta 18), reused 22 (delta 16), pack-reused 13397 (from 3)
Receiving objects: 100% (13429/13429), 106.38 MiB | 37.36 MiB/s, done.
Resolving deltas: 100% (7291/7291), done.
Filtering content: 100% (45/45), 69.03 MiB | 45.25 MiB/s, done.
```

```
# Navigate to repository and install
%cd lerobot
```

```
/content/lerobot
```

```
# Install LeRobot core package
!pip install .
```

Show hidden output

```
# Install PushT environment and dependencies
!pip install 'lerobot[pusht]' hydra-core
```

Show hidden output

```
# Download pretrained weights
!curl 'https://drive.usercontent.google.com/download?id=1wC2rVCrM0U33tJZH_Ai4WqKfywXnuzg5&export=download&authuser=0&confirm=t&uuiid
-H 'User-Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64; rv:144.0) Gecko/20100101 Firefox/144.0' \
-H 'Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8' \
-H 'Accept-Language: en-US,en;q=0.5' \
-H 'Accept-Encoding: gzip, deflate, br, zstd' \
-H 'Connection: keep-alive' \
-H 'Cookie: NID=525=hiKeUKfvrVpNAQeRlVzEny-6DH_A8G5MdDTko_YMZG2t3ZZQySwW9zo1j9dAhPJmL2EuuoVwAjCVhD97jHMjWY9UHK1y9B21mUBbA_ENo9S8g
-H 'Upgrade-Insecure-Requests: 1' \
-H 'Sec-Fetch-Dest: iframe' \
-H 'Sec-Fetch-Mode: navigate' \
-H 'Sec-Fetch-Site: same-site' \
-H 'Priority: u=4' \
-H 'TE: trailers' \
--output /content/lerobot_dp_checkpoints.tar.gz
```

% Total	% Received	% Xferd	Average Speed	Time	Time	Time	Current
			Dload Upload	Total	Spent	Left	Speed
100	934M	100	934M	0	0	58.0M	0
				0:00:16	0:00:16	--:--:--	70.7M

```
# Unzip
!tar -xvzf /content/lerobot_dp_checkpoints.tar.gz
```

```
./diffusion_pusht_migrated/
./diffusion_pusht_migrated/policy_preprocessor.json
./diffusion_pusht_migrated/model.safetensors
./diffusion_pusht_migrated/policy_preprocessor_step_3_normalizer_processor.safetensors
./diffusion_pusht_migrated/README.md
./diffusion_pusht_migrated/policy_postprocessor_step_0_unnormalizer_processor.safetensors
./diffusion_pusht_migrated/policy_postprocessor.json
./diffusion_pusht_migrated/config.json
```

Part 2: Training a Diffusion Policy

Training Procedure

The training process follows the standard supervised learning paradigm for imitation learning:

Dataset: Expert demonstrations $\mathcal{D} = \{(\mathbf{o}_i, \mathbf{a}_i)\}_{i=1}^N$

- Observations: $\mathbf{o}_i = \{I_i, s_i\}$ (RGB images and proprioceptive state)
- Actions: $\mathbf{a}_i \in \mathbb{R}^{T_a \times d_a}$ (action sequences)

Training Loop:

```
For each batch {(o_i, a_i)}:
1. Sample diffusion timestep:  $t \sim \text{Uniform}(1, T)$ 
2. Sample noise:  $\epsilon \sim \mathcal{N}(0, I)$ 
3. Add noise to actions:  $\mathbf{a}_t = \sqrt{\alpha_t} \cdot \mathbf{a}_0 + \sqrt{1 - \alpha_t} \cdot \epsilon$ 
4. Predict noise:  $\epsilon_{\text{pred}} = \epsilon_\theta(\mathbf{a}_t, \mathbf{o}_i, t)$ 
5. Compute loss:  $L = \|\epsilon - \epsilon_{\text{pred}}\|^2$ 
6. Update  $\theta$  via gradient descent
```

Hyperparameter Configuration

Key training hyperparameters:

Diffusion Schedule:

- Number of diffusion steps: $T = 100$
- Noise schedule: Linear or cosine β_t schedule
- Variance type: Fixed or learned

Network Architecture:

- Vision encoder: U-net

1. Input:

- Shape: (B, 2, 16)

2. Encoder:

- Enc Block 0: 2 ResBlocks $\rightarrow$ (B, 512, 16)
- Downsample 0: Conv1d (stride=2) $\rightarrow$ (B, 512, 8)
- Enc Block 1: 2 ResBlocks $\rightarrow$ (B, 1024, 8)
- Downsample 1: Conv1d (stride=2) $\rightarrow$ (B, 1024, 4)
- Enc Block 2: 2 ResBlocks $\rightarrow$ (B, 2048, 4)

3. Bottleneck:

- 2 ResBlocks $\rightarrow$ (B, 2048, 4)

4. Decoder:

- Dec Block 0: 2 ResBlocks $\rightarrow$ (B, 1024, 4)
- Upsample 0: ConvTranspose1d $\rightarrow$ (B, 1024, 8)
- Dec Block 1: 2 ResBlocks $\rightarrow$ (B, 512, 8)
- Upsample 1: ConvTranspose1d $\rightarrow$ (B, 512, 16)
- Dec Block 2: 2 ResBlocks $\rightarrow$ (B, 512, 16)

5. Output:

- Final Conv: Conv blocks $\rightarrow$ (B, 2, 16)

Optimization:

- Optimizer: AdamW
- Learning rate: 1×10^{-4} to 3×10^{-4}
- Batch size: 64-256
- Weight decay: 1×10^{-6}

Action Prediction:

- Prediction horizon T_p : 16 timesteps
- Observation horizon T_o : 2 timesteps
- Action execution T_e : 8 timesteps

Implementation Details

The training command below specifies:

- `policy.type=diffusion`: Diffusion policy
- `env.type=pusht`: PushT manipulation environment
- `eval_freq=10000`: Evaluate policy every 10,000 training steps
- `log_freq=200`: Log training metrics every 200 steps
- `save_freq=25000`: The frequency of steps to save checkpoint at
- `wandb.enable=false`: Disable Weights & Biases experiment tracking

Computational Requirements

Training Time: Approximately 12 hours on a single NVIDIA V100 GPU for 200K steps

Hardware Requirements:

- GPU: NVIDIA GPU with ≥ 16 GB VRAM recommended
- RAM: ≥ 32 GB system memory
- Storage: ~ 10 GB for datasets and checkpoints

Training from Scratch

Note: The cell below trains a new model from scratch. This process is computationally intensive. For rapid experimentation, you may prefer to use pre-trained checkpoints (see Part 3).

```
# Train
%env WANDB_SERVICE=
!PYTHONPATH=/content/lerobot python src/lerobot/scripts/lerobot_train.py \
  --policy.type=diffusion \
  --env.type=pusht \
  --dataset.repo_id=lerobot/pusht \
  --policy.push_to_hub=false \
```

```
--dataset.image_transforms.enable=true \  
--eval_freq=10000 \  
--log_freq=200 \  
--save_freq=25000 \  
--wandb.enable=false \  
--output_dir=outputs/train/diffusion_pusht
```

▼ Part 3: Policy Evaluation

Evaluation Protocol

We assess the trained policy using standard metrics:

Success Rate:

$$\text{Success Rate} = \frac{1}{N} \sum_{i=1}^N 1[\text{task}_i \text{ completed}]$$

Average Return:

$$J(\pi) = \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^T r_t \right]$$

Model Checkpoint Requirements

The checkpoint directory must contain **six required files**:

1. `model.safetensors` - Model weights
2. `config.json` - JSON configuration
3. `policy_postprocessor_<step>.safetensors`
4. `policy_preprocessor.json`
5. `policy_preprocessor_<step>.safetensors`
6. `policy_postprocessor.json`

Example directory structure:

```
/content/drive/MyDrive/Lerobot_DP_Checkpoints/  
├─ model.safetensors  
├─ config.json  
├─ policy_postprocessor.json  
├─ policy_preprocessor.json  
├─ policy_postprocessor_step_0_unnormalizer_processor.safetensors  
└─ policy_preprocessor_step_3_normalizer_processor.safetensors
```

```
# Set the checkpoint directory path  
# This should contain: model.safetensors, config.json, config.yaml  
model_file = '/content/lerobot/diffusion_pusht_migrated'  
  
# Run evaluation  
!PYTHONPATH=/content/lerobot python src/lerobot/scripts/lerobot_eval.py \  
  --policy.path={model_file} \  
  --env.type=pusht \  
  --eval.n_episodes=10 \  
  --eval.batch_size=1 \  
  --policy.device=cuda \  
  --output_dir=./eval_output
```

[Show hidden output](#)

▼ Part 4: Analyzing the Denoising Process

Visualization and Interpretation

Understanding the iterative denoising mechanism is crucial for interpreting how Diffusion Policy generates actions.

The Denoising Trajectory

At inference time, actions evolve through T denoising steps:

$$\mathbf{a}^{(T)} \xrightarrow{\epsilon_\theta} \mathbf{a}^{(T-1)} \xrightarrow{\epsilon_\theta} \dots \xrightarrow{\epsilon_\theta} \mathbf{a}^{(1)} \xrightarrow{\epsilon_\theta} \mathbf{a}^{(0)}$$

Key Observations:

- Early steps ($k \approx T$): Large-scale structure emerges (coarse trajectory shape)
- Middle steps ($k \approx T/2$): Refinement of motion primitives
- Late steps ($k \approx 0$): Fine-grained adjustments and smoothing

Action Chunking Strategy

Prediction vs. Execution Horizons:

Given observation $\mathbf{o}_t$, the policy predicts:

$$\hat{\mathbf{a}}_{t:t+15} = \pi_\theta(\mathbf{o}_t) \in \mathbb{R}^{16 \times d_a}$$

But only executes:

$$\mathbf{a}_{t:t+7} = \hat{\mathbf{a}}_{t:t+7}$$

Theoretical Justification:

1. **Receding Horizon Optimality:** At each step, we solve:

$$\max_{\mathbf{a}_{t:t+T_p-1}} \mathbb{E}[R(\tau) | \mathbf{o}_t]$$

But execute only $T_e < T_p$ actions, then re-optimize.

2. **Temporal Consistency:** The overlap creates an implicit regularization:

$$\mathbf{a}_{t+T_e:t+T_p-1}^{(i)} \approx \mathbf{a}_{t+T_e:t+T_p-1}^{(i+1)}$$

where superscript (i) denotes prediction iteration.

3. **Empirical Performance:** Chi et al. (2023) report:

- $T_p = T_e$: 70% success rate
- $T_p = 16, T_e = 8$: 85% success rate (+15% improvement)

Visualization Configuration

The evaluation command with visualization enabled:

```
python lerobot/scripts/eval.py \
  -p outputs/train/diffusion_pusht/checkpoints/005000/pretrained_model \
  eval.n_episodes=10 \
  eval.vis_noise=True
```

Visualization Parameters:

- `eval.n_episodes`: Number of rollout episodes
- `eval.vis_noise`: Visualize denoising trajectory
- Display last 20 denoising steps for computational efficiency

Interpreting Results

What to observe:

1. **Action manifold learning:** Do denoised actions lie on a lower-dimensional manifold?
2. **Multimodality:** Can the policy generate diverse solutions from different noise initializations?
3. **Smoothness:** Are generated trajectories C^1 or C^2 continuous?
4. **Task-relevant structure:** How quickly do task-specific patterns emerge during denoising?

Theoretical Insight: Score-Based Perspective

From the score-based generative model viewpoint, the noise predictor approximates:

$$\epsilon_\theta(\mathbf{a}_t, \mathbf{o}, t) \approx -\sqrt{1 - \bar{\alpha}_t} \nabla_{\mathbf{a}_t} \log p_t(\mathbf{a}_t | \mathbf{o})$$

This score function points toward regions of high probability in the conditional action distribution $p(\mathbf{a} | \mathbf{o})$, effectively implementing gradient ascent in probability space.

```
# Run evaluation WITH denoising visualization
# Add --eval.vis_noise=True to enable visualization
```

```
!PYTHONPATH=/content/lerobot python src/lerobot/scripts/lerobot_eval.py \
--policy.path={model_file} \
--env.type=pusht \
--eval.n_episodes=3 \
--eval.batch_size=1 \
--eval.vis_noise=True \
--policy.device=cuda \
--output_dir=./eval_output_with_viz

# Videos will be saved to: ./eval_output_with_viz/videos/
# Each video shows the last 20 denoising steps overlaid on environment frames
# Color-coded (plasma): purple (early actions) → yellow (later actions)
```

Show hidden output

```
from IPython.display import HTML
from base64 import b64encode

mp4 = open("./eval_output_with_viz/videos/pusht_0/eval_episode_0.mp4", 'rb').read()
data_url = "data:video/mp4;base64," + b64encode(mp4).decode()
HTML(f'<video width=640 controls><source src="{data_url}" type="video/mp4"></video>')
```

1:07 / 1:07



Visualization Implementation Details

The `render_diffusion_frame()` callback in `lerobot_eval.py`:

1. Captures base environment rendering
2. Extracts last 20 denoising steps from policy output
3. Overlays action predictions as scatter points (action space $\rightarrow$ pixel space)
4. Creates one frame per denoising timestep

Key transformations:

```
# Scale actions from action space to pixel coordinates
scale_width = image_width / action_space_width
scale_height = image_height / action_space_height
scaled_actions = actions * [scale_width, scale_height]
```

```
# Color progression: plasma colormap
colors = plt.cm.plasma(np.linspace(0, 1, num_actions))
```

Performance notes:

- Requires `(batch_size=1)` (single environment)
- Significantly slower due to matplotlib rendering at each step (~3-5 seconds per environment step)
- Video frame count: $\approx (n_steps / 8) \times 20 + n_steps$ (20 denoising frames per action chunk)

Part 5: Summary

Key Contributions

1. Action Diffusion Framework:

- Reformulates action prediction as iterative denoising
- Enables multimodal action distributions
- Provides implicit temporal smoothness

2. Architectural Innovations:

- Transformer-based denoising with cross-attention to observations
- Receding horizon action chunking with temporal overlap
- Vision-language-action co-training (recent variants)

3. Empirical Performance:

- State-of-the-art on manipulation benchmarks
- Superior sample efficiency vs. reinforcement learning
- Improved generalization vs. behavioral cloning

Theoretical Analysis

Expressiveness: Diffusion models approximate arbitrary continuous distributions:

$$p_{\theta}(\mathbf{a}|\mathbf{o}) \xrightarrow{T \rightarrow \infty} p_{\text{data}}(\mathbf{a}|\mathbf{o})$$

Sample Complexity: Compared to behavioral cloning:

- **BC:** Assumes unimodal $p(\mathbf{a}|\mathbf{o}) = \mathcal{N}(\mu_{\theta}(\mathbf{o}), \Sigma)$
- **Diffusion:** Models full distribution without parametric assumptions